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Case Reports

Acute Chagas' disease in postrenal transplant and treatment with benzonidazole☆

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Abstract

Transplanted organs may act as a route of transmission of infectious diseases, such as Chagas' disease. The aim of this study was to describe the transmission of the *Trypanosoma cruzi* through a renal transplant and the anatomo-clinical evolution of the patient after treatment with benzonidazole. The patient was a 31-year-old white male from the State of Minas Gerais in Brazil. He had renal failure secondary to diabetes and later received a kidney from a cadaveric donor. The patient was undergoing immunosuppression therapy with azathioprine, cyclosporine A, and prednisone. After the transplant, he developed an acute phase of Chagas' disease and complications from diabetes and died 2 months later. In the autopsy, *T cruzi* amastigotes were found in the transplanted kidney, heart, bladder, liver, and pancreas. An important reduction in the parasitemia was obtained through the treatment of the infection with benzonidazole; however, the patient died due to complications from diabetes associated with tissue lesions caused by *T cruzi*.

Keywords: Benzonidazole; Chagas' disease; Renal transplant; *Trypanosoma cruzi*

1. Introduction

In endemic areas, *Trypanosoma cruzi* is transmitted by triatomines, but in urban centers, transmission occurs mainly through blood transfusions and organ transplants. Migration of infected people from rural areas to cities in developed and developing countries results in these individuals becoming potential blood and organ donors [1].

After renal transplants, patients from endemic areas may develop the acute phase of the disease as a consequence of reactivation or transmission from grafting or blood transfusion [2]. The transmission of the disease through organ transplant occurs due to parasites in the tissue. This form of transmission and development of the acute Chagas' disease has been described by other authors [3-5]. In renal transplant patients, the specific treatment of Chagas' disease is effective for a short time, but there is not yet sufficient experience to outline subsequent implications. Moreover, the immunosuppression therapy is well tolerated in association with benzonidazole [6].

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The aim of this case report was to describe the transmission of the *T cruzi* through a renal transplant and the anatomo-clinical evolution of the patient after treatment with benzonidazole.

2. Case

The patient was a 31-year-old single white male from the State of Minas Gerais, Brazil. The patient developed chronic renal failure secondary to diabetes and underwent hemodialysis for 6 years. The patient was submitted to a kidney transplant from a nonrelated cadaveric donor, whose serologic status for *T cruzi* was unknown at the time of the transplant. Later, a positive serology was observed for Chagas' disease (complement fixation reaction and indirect hemagglutination). In pretransplant laboratory investigations, the receptor presented negative serology for human immunodeficiency virus, cytomegalovirus, hepatitis C, Chagas' disease, and a positive serology to hepatitis B. In the immediate postsurgery, the patient presented a favorable evolution with satisfactory urinary output. The immunosuppressed therapy was done with cyclosporine A (150 mg/d), azathioprine (200 mg/d), and prednisone (60 mg/d). On the sixth day postsurgery, daily febrile peaks were observed, followed by oliguria, painful abdominal palpation in the transplant region and drainage of serosanguinous secretions from the surgical wound. Abdominal ultrasound examination revealed images suggesting severe pyeloureteral dilation and satisfactory renal perfusion with elevated resistance in the transplanted kidney. Because of a progressive loss of renal functions, the patient was submitted to an exploratory

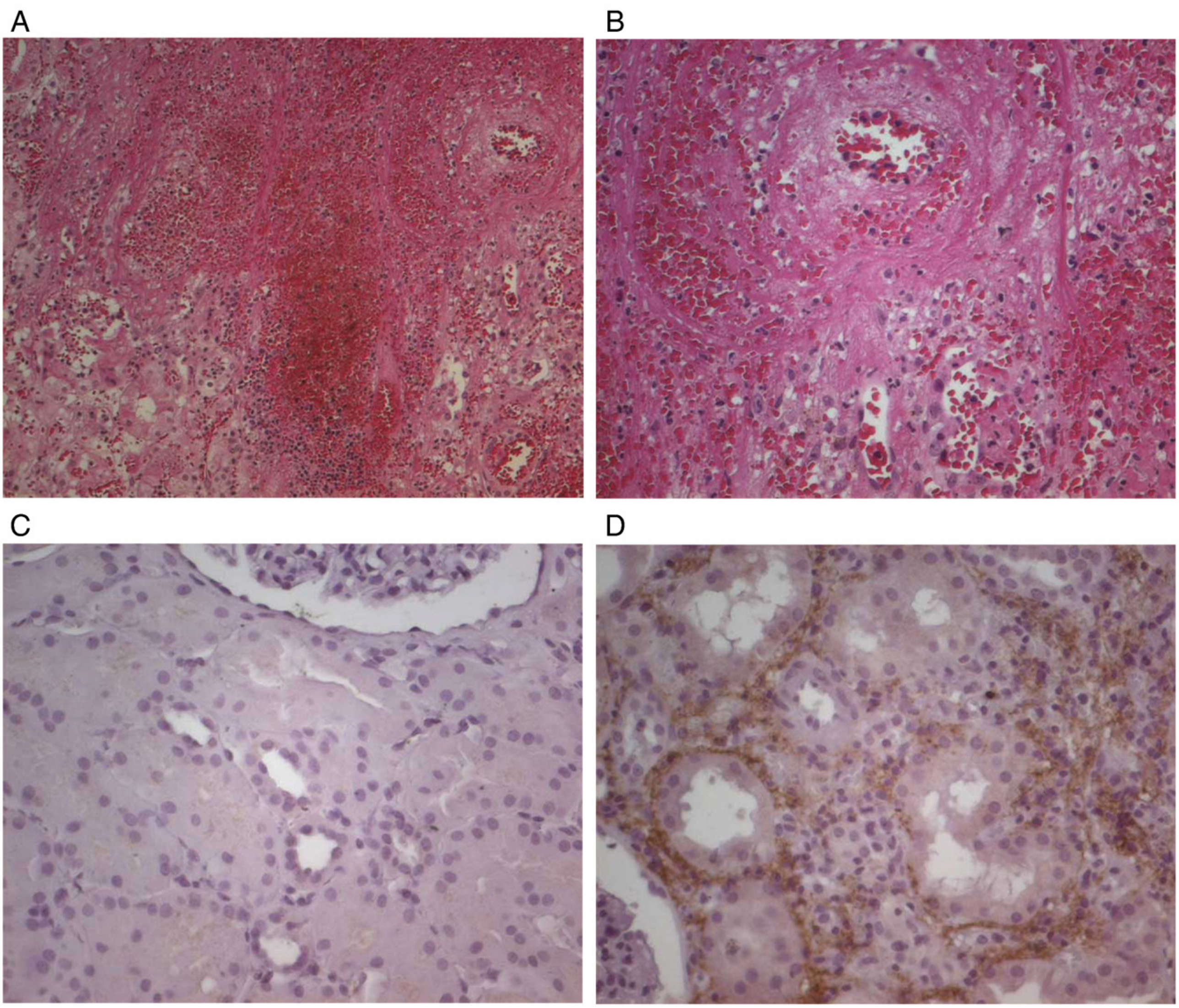

Fig. 1. Micrographies from kidney biopsy with acute rejection, type III, show focus hemorrhagic and arteritis "transmural" (A, "low power"; B, "high power," hematoxylin-eosin) and micrographies from kidney biopsy, made at the time of transplantantion, biopsy zero, C4dzero = negative (C, "high power," anti-C4d immunohistochemistry), and kidney biopsy on the 11th day postsurgery with positive staining for C4d in more than 50% of peritubular capillaries = positive (D, "high power," anti-C4d immunohistochemistry).

Table 1
Quantification of parasites in peritoneal liquid and blood using Brenner method before and during benzonidazole therapy

Day of the postsurgery	Peritoneal liquid (trypomastigotes/0.1 mL)	Blood (trypomastigotes/0.1 mL)
57th	770 000	–
59th	–	232 000
61st	–	114 000
63rd	–	35 000

laparotomy and presented a moderate rupture of the upper renal pole with a large perihilar hematoma. A biopsy of the graft was performed on the 11th day postsurgery, showed acute antibody-mediated rejection according to the Banff classification [7]. There were 20 with glomeruli, 5 of them were totally sclerosed, and others were with few lymphocytes or polymorphonuclears. The tubules and interstitium presented an extensive area of bleeding that was destroying the renal parenchyma, foci of coagulative necrosis, and diffuse mononuclear inflammatory infiltrate (i3) with moderate tubulitis (t2). Some arterioles showed hyaline thrombi. There was vasculitis in the arteries, and in one of them, the infiltrate became transmural. In immunohistochemistry for the detection of C4d, a blade from the paraffined reserve material was used. The primary anti-C4d antibody (1:600, ALPCO, Windham, NH, USA) was used. According to the percentage of area marked in peritubular capillaries, C4d3 was considered to be diffuse because the percentage of area marked in peritubular capillaries was more than 50% (meaning and interpretation: positive) (Fig. 1). Later, the patient was submitted to 3 cycles of methylprednisolone pulse therapy (0.5-3.0 g/pulse) and evolved with satisfactory diuresis but maintained tachycardia and fever. Subsequently, a decrease of the renal functions was observed, and hemodialysis was resumed. On the 57th day of the postsurgery, *T cruzi* trypomastigotes were found in the peripheral blood smears of a routine hemogram, and the acute phase of Chagas' disease was diagnosed. The peritoneal liquid was examined, and 770 000 trypomastigotes/0.1 mL were detected. Immediate treatment with benzonidazole (5 mg/kg per day) was started, the use of azathioprine and cyclosporine A was suspended, and the use of prednisone (10 mg/d) was maintained with gradual reduction of the dosage. The parasite burden measured on the 59th was quantified at

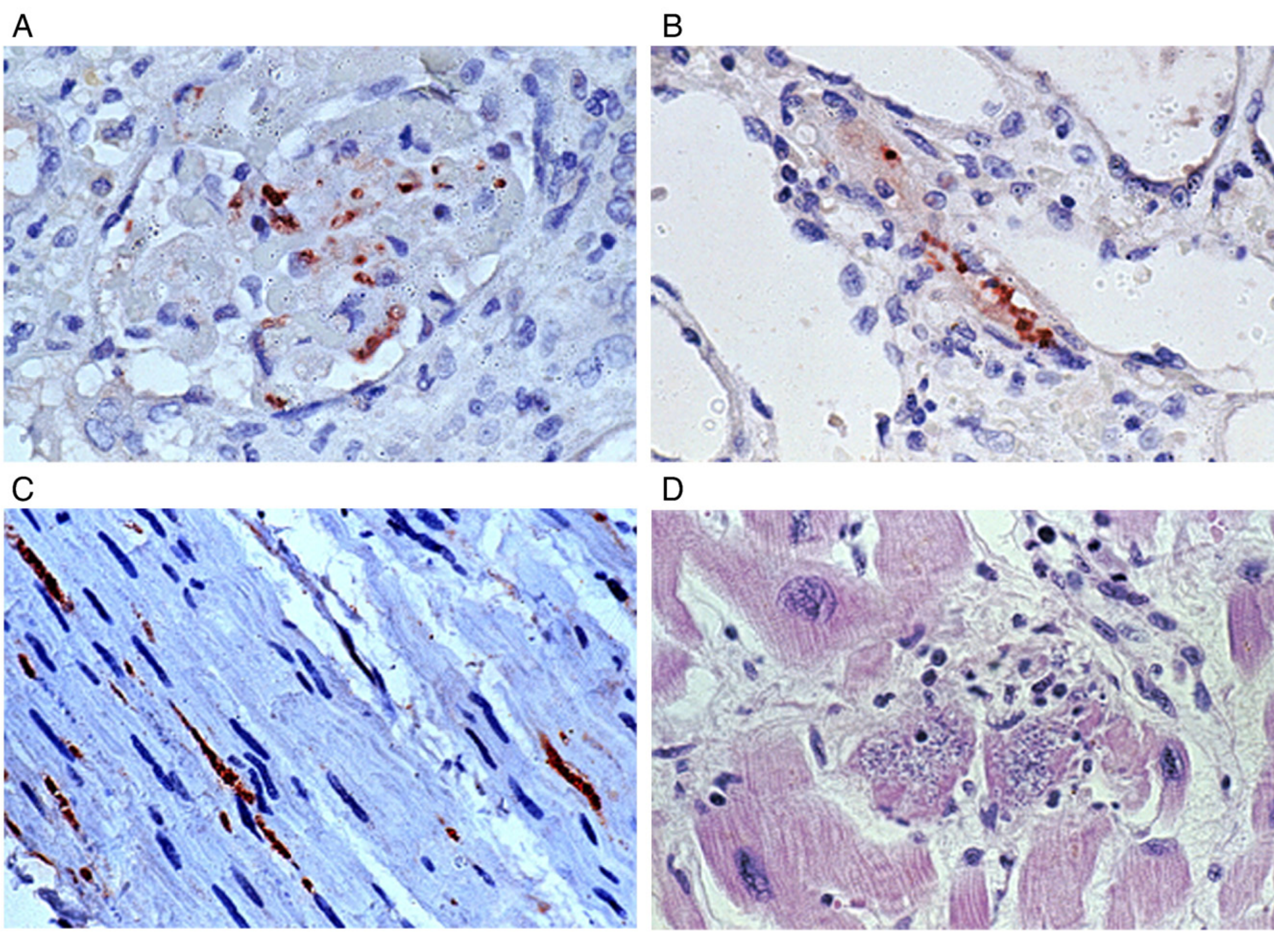

Fig. 2. Micrographies from autopsy of a patient with acute Chagas' disease after kidney transplant showing kidney (A and B) and bladder (C) with positive staining for *T cruzi* antigens (anti-*T cruzi*, immunohistochemistry) in the glomerulus (A, "high power"), renal interstitium (B, "high power"), and in the smooth muscle of bladder wall (C, "high power"), and nests of *T cruzi* in myocardiocytes (D, "high power").

232 000 trypomastigotes/0.1 mL (Table 1). The parasitic kinetics can be observed on Table 1. The quantitative analysis of the samples was handled by the Brenner method (Brenner, 1963) and the qualitative analysis by the microhematocrit method. The investigation of the parasite in the cerebrospinal fluid was negative.

No alterations suggesting parasitism were found in the routinely stained cuts made from the biopsies of the transplanted kidney ("zero" and on the 11th day postsurgery). The biopsy "zero" showed that there were 7 glomeruli, 3 of them were totally sclerosed, and the others were normal. The tubules and interstitium showed mild to moderate degenerative and necrotic changes in tubular epithelial cells. Blood vessels, arterioles, and interlobular arteries were within normal standards. Reviews of the biopsy were done using immunohistochemistry (peroxidase-immunoperoxidase technique for *T cruzi*), and parasitism was not observed in either biopsy. On the other hand, the transplanted kidney fragments obtained from the autopsy were positive for *T cruzi* antigens on the immunohistochemistry analyses (Fig. 2).

The specific treatment of Chagas' disease developed favorably, with improvements in the febrile peaks, level of conscientiousness, and a significant decrease in the parasitemia. During the period of internment, the patient presented difficulty in controlling the glycemic levels. On the 63rd, the patient's clinical state worsened with weakness, asthenia, inappetence, tachycardia, and hypoglycemia. He died the following day, which was the seventh day after adhering to the treatment with benzonidazole.

In the autopsy, terminal native kidneys (60 g each) with alterations suggesting diabetic nephropathy containing Kimmelstiel-Wilson lesions were observed. The transplanted kidney (250 g) had a tigroid external surface and microscopic evidence of chronic nephropathy of the graft, typical of chronic vascular rejection with overlapping of acute rejection and tubulointerstitial nephritis with infection from the *T cruzi*. Intense parasitism was observed in the heart, bladder, spleen, liver, and pancreas. Signs of myelotoxicity were found in the bone marrow. No parasites were found in the central nervous system. Moreover, in the heart, acute chagasic myocardiopathy associated with hypertensive cardiopathy was observed. The organ weighed 434 g, and the ratio of heart weight to body weight was 0.82%.

3. Discussion

A case of transmission of Chagas' disease by renal transplant was presented confirming the importance of this via of transmission in urban centers and endemic areas for this disease as described by some authors [1,3,8]. Normally, patients infected via this form present seroconversion ranging from a few days to several months. Some patients do not present trypanosomes in the parasite investigation, making it difficult to establish a differential diagnosis with rejection [1,3]. In this case, the patient presented parasitemia that facilitated the diagnosis and immediate treatment with benzonidazole.

The signs and symptoms of the acute phase of the disease are as follows: fever, vomiting, diarrhea, edemas of face and extremities, lymphadenopathy, hepatosplenomegaly, tachycardia, meningism, and seizures [9-11]. This phase, specially indicated by fever, also makes the differential diagnosis with rejection more difficult [6]. Parasites can also be found in cells of the neuroglia, mononuclear-macrophage system, heart, and gastrointestinal tract. Diagnosis is based on the detection of the parasites in the peripheral blood, specific serologies for *T cruzi*, and currently through molecular methods. Because of treatment in symptomatic cases or development of a specific immune response to the parasite, the symptoms decline at around 2 months after infection, and the patients can remain asymptomatic or develop an undetermined form of the disease [10].

Some transplant patients in the chronic phase of the disease present parasitemia, whereas others do not, even when these patients do not receive chemoprophylaxis and are submitted to immunosuppression treatment [9,10]. This has led some authors to conclude that drug immunosuppression would not significantly affect the evolution of the disease [12]. Nevertheless, it might be asked if the immunosuppression therapy could lead to an acute phase or even the reactivation of the disease [3]. Some variables that can be related to clinical presentation are age, parasite burden, virulence, and pathogenicity of the strain, tropism, and route of transmission of the parasite [8]. In addition, alterations of the inherent immune response for the underlying disease can, in association with the immunosuppression therapy, predispose for opportunistic diseases [13]. Reactivation of the Chagas' disease in transplanted patients is generally more common in patients submitted to heart transplant, possibly as a result of the differences in the immunosuppression therapy for these patients.

Another difficulty in endemic areas is to separate cases of acute infection through transplanted organs, from those resulting from contaminated blood [8] or from the vector [10]. The possibility of reactivation of the disease in chronically infected patients, even when these patients initially present negative serology, must be considered.

Cantarovich et al (1991) [9] suggested that the kidneys were unaffected by the parasitism from the *T cruzi* because receptors with negative serology for Chagas' disease did not present signs of infection or parasitemia by xenodiagnosis after receiving kidneys from donors with positive serology to the disease. Subsequently, it was observed that kidneys could house parasite infected cells from the *T cruzi* and thus, constitute a source of infection [14].

T cruzi amastigotes have been observed in the brain of patients submitted to renal transplant and undergoing immunosuppression therapy [15]. In another case, analysis of the cerebrospinal fluid presented trypomastigotes [13]. In this case, no parasites were observed in the brain cuts analyzed under the microscope.

In this patient studied, parasitism was intense in the bladder, discreet in the liver, spleen, and pancreas. The heart presented acute Chagas' myocarditis with nests of amastigotes. Furthermore, there was an intense parasitism in the peritoneal liquid, which could be explained by tropism by regional macrophages, such as those observed in mice intraperitoneally inoculated with the Y strain of the *T cruzi* [16,17]. Moreover, there was, in the evolution of the patient, an intense response of the *T cruzi* to the benzonidazole, with consequent reduction in the sanguineous parasitism during the 7 days of treatment, associated with a clinical improvement in the acute infection. Nevertheless, the patient died due to complications from diabetes associated with tissue lesions caused by *T cruzi*. In endemic areas for Chagas' disease, it is important to be aware of its transmission via grafting and the reactivation of the disease in receptors with a positive serology.

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